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MILITARY HANDBOOK

TOWING NONSELF-PROPELLED FLOATING STRUCTURES

AMSC N/A

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ABSTRACT

This handbook presents guidance and general criteria for use in preparing tows of nonself-propelled structures in inland and ocean waters. It is intended for use by qualified Naval and civilian personnel. Included are data on towing resistance calculations, planning and methods, crew provisions, ground tackle, towline choice and fittings, checklists, outfitting, and recordkeeping.

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FOREWORD

This handbook has been developed from an evaluation of facilities in the shore establishment, from surveys of the availability of new materials and construction methods, and from selection of the best design practices of the Naval Facilities Engineering Command (NAVFACENGCOM), other Government agencies, and the private sector. This handbook was prepared using, to the maximum extent feasible, national professional society, association, and institute standards. Deviations from this criteria, in the planning, engineering, design, and construction of Naval shore facilities, cannot be made without prior approval of NAVFACENGCOM HQ Code 04.

Design cannot remain static any more than can the functions it serves or the technologies it uses. Accordingly, recommendations for improvement are encouraged and should be furnished to Commander, Western Division, Code 406C, Naval Facilities Engineering Command, P.O. Box 727, San Bruno, CA 94066-0720, telephone (415) 877-7422.

THIS HANDBOOK SHALL NOT BE USED AS A REFERENCE DOCUMENT FOR PROCUREMENT OF FACILITIES CONSTRUCTION. IT IS TO BE USED IN THE PURCHASE OF FACILITIES ENGINEERING STUDIES AND DESIGN (FINAL PLANS, SPECIFICATIONS, AND COST ESTIMATES). DO NOT REFERENCE IT IN MILITARY OR FEDERAL SPECIFICATIONS OR OTHER PROCUREMENT DOCUMENTS.

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WEIGHT-HAULING EQUIPMENT AND SERVICE CRAFT CRITERIA MANUALS

Criteria Manual	Title	PA
DM-38.01	Weight Handling Equipment and Service Craft	HDQTRS
DM-38.02	Dredging Equipment	SOUTHDIV
MIL-HDBK-1038/3	Yard Craft	WESTDIV
DM-38.4	Pile Driving Equipment	LANTDIV
MIL-HDBK-1038/5	Towing Nonself-propelled Floating Structures	WESTDIV

NOTE: Design manuals, when revised, will be converted to military handbooks and listed in the military handbook section of NAVFAC P-34.

This handbook is issued to provide immediate guidance to the user. However, it may or may not conform to format requirements of MIL-HDBK-1006/3 and will be corrected on the next update.

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TOWING NONSELF-PROPELLED FLOATING STRUCTURES

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Section 1: INTRODUCTION

1.1 Scope. This handbook is a guide for the towing of floating craft that fall within the technical responsibility of the Naval Facilities Engineering Command (NAVFACENGCOM). It presents guidance for planning and preparing tows and selecting system components for tow of nonself-propelled service craft and floating structures in astern towing operations to minimize damage or loss of craft in the open sea, rivers, and harbors. It also includes descriptions of typical units to be towed, towing arrangements, gear design and component selection, special outfitting equipment and towing vessel considerations upon which a sound towing operation may be formulated.

1.2 Cancellation. This handbook, cancels NAVFAC DM-38.5, Towing Nonself-Propelled Floating Structures, March 1981.

1.3 References. A number enclosed with ( ) indicates references at the end of this handbook, except for equations. Equations are also cited in the text, but will always be preceded by the word "Equation."

NAVSHIPS 0925-000-1000, The U.S. Navy Towing Manual, Vol. I and II, 1971, (1) provides a thorough source of guidance and data for operation and design of towing systems. It is strongly recommended that these manuals and their pending updated versions be used as a supplement to this handbook.

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Section 2: PLANNING TOWING EXPEDITIONS

2.1 Introduction. A towing operation is begun when the Commanding Officer (CO) of a specific craft issues a directive which specifies the new station to which the craft will be towed and including any special including the effects of changing course, speed, and sea conditions is the responsibility of the operating forces. The CO of a craft will be primarily concerned with the preparation of the craft for towing. A task force organization with centralized responsibility, is recommended for implementing the towing operations indicated herein. Special towing problems, with all pertinent facts may be referred to this WESTDIV NAVFACENGCOM for review and recommendation.

Towing expeditions should be planned to take advantage of the best routes and weather conditions. There should be no need to enter any port subject to emergencies, such as damage sufficient for the CO to declare the tow unfit for further ocean tow, or scheduled port calls. If a unit is damaged or disabled, further towing will be dependent on the judgment of the Officer-In-Charge (OIC) of the tow.

2.2 Unmanned Tows. Under normal conditions, and after proper securing for sea, most tows can be undertaken without a riding crew. Barges, dump scows, carfloats, floating derricks, dredges, pontoons, cranes, pile drivers, dry docks, caissons, auxiliary and combatant ships can be towed without a riding crew. Any hull considered seaworthy can be taken unmanned under tow. A seaworthy hull has watertight security, structural soundness, proper position of the centers of gravity and buoyancy, good stability characteristics, and security of all cargo and equipment. Hull condition is described in more detail in NAVFAC MIL-HDBK-1038/3, Yard Craft (2). Unmanned tows may require frequent stops for boarding of towed unit by towing vessel personnel to accomplish inspection and other purposes.

2.3 Riding Crews. OPNAVINST 4740.1C, Ocean Tows (3), specifies that riding crews will be provided on any ocean tow only as directed by the Chief of Naval Operations (CNO), the commander whose mission requires the performance of the tow, or the command ordering accomplishment of the tow. When required, the number of personnel in a riding crew shall be limited to those required for maintenance and security during the voyage. A suggested towing crew complement would be one officer and enlisted ratings such as signalmen, seamen, pharmacist's mate, electricians, motor machinist, ship's cook, and boatswain's mates. The size and type of the towed craft, its commensurate value, length of tow, foreseeable ocean environmental factors, and feasibility of providing safe riding crew accommodations should be considered. Riding crew duties include handling of lines, checking and reporting conditions of tows, standing fire and flooding watch, and applying necessary safety and damage control measures.

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2.4 Towing Outfitting Allowances. Equipment and gear needed for towing have been established and are usually included in the Equipment Allowance Lists for any particular craft at the time of procurement. Detail drawings showing arrangement of gear and special preparations for towing are contained When craft allowance tackle is not available, the guidance for towing gear arrangements and selection will indicate suitable components for the towing operation (1). When craft allowance tackle is not available, the guidance for towing gear arrangements and selection hereinafter and in Reference 1 will indicate suitable components for the towing operation. Duplicate towing gear often is provided on the after end of important tows to insure against an emergency involving failure or loss of the forward gear.

2.5 Cargo Carrying. Most floating structures may carry cargo while under tow. When cargo is stowed aboard, the limiting conditions of structural strength and floating stability of the towed unit must be considered unless the units are designed to carry the specified cargo loads. Loading normally presents no particular problems when towing lighter barges and pontoon structures of various assemblies other than considerations of towing freeboards, physical security, and protection from environmental effects. All cargo must be adequately secured to resist the dynamic forces caused by rolling and pitching of the towed unit in a seaway. Some craft may ship considerable water over their decks in a rough sea, thus limiting the amount and type of cargo they may carry and may require installation of wave bulwarks on the bow and/or stern. A dredge, crane, derrick, pile driver, or similar special structure should be loaded in accordance with relevant plans. When plans are not available, consult NAVFACENGCOM, Code 04A3, concerning limiting conditions and advice on loading and securing methods.

2.6 Damage Control. A towed unit may be damaged by combat action, structure collision, storms, or running aground. The riding crew, or the towing vessel command, should be thoroughly familiar with the subdivisions of the hull and the service systems, and be instructed in damage control procedures for the towed structure. Damage control books have been prepared for most military-type craft and should be kept aboard. If manuals are not available, general damage control knowledge and experience of the OIC shall determine actions to be taken.

2.7 Hearing Conservation. Impact sound pressure levels from equipment used on barges can be expected to exceed 140 dB. Unprotected personnel exposed to these high impact sound pressure levels for extended periods of time may incur permanent hearing loss. Therefore, it is recommended that a qualified industrial hygienist be consulted to prescribe the appropriate degree of hearing protection necessary to preserve hearing.

2.8 Safety Precautions. Principals of good seamanship should always be followed as the last safety guide. Basic safety precautions are presented in OPNAVINST 5100.19A, Article 0212, paras. 9 and 10, and Article 0281, Navy Safety Precautions for Forces Afloat (9). It also presents general safety precautions for towing. These general safety precautions are reiterated in paras. 2.8.1 through 2.8.9.

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2.8.1 General Safety. When synthetic rope is used in towing operations without a towing engine, the towing hawser must have a scope of at least 200 fathoms. The hawser shall never be sprung so much that it straightens out clear of the water, and shall never be allowed to drag on the bottom. Steps shall be taken to reduce chafing and wear on the towline. Provide proper chafing gear, and freshen the nip everyday if chafing chains are not used. Ensure the material condition of the towline, bridle and allied towing gear is in good condition. Never use a hawser that is kinked or badly frayed. Always rig properly and perform adequate maintenance. Towing hawsers shall be inspected thoroughly during overhaul periods. The hawsers shall be proof-tested if inspection reveals any unsatisfactory conditions.

2.8.2 Towing Precautions. Before disconnecting the anchor chain on the forecastle for towing, inserting a swivel, mooring to a buoy, or rousing out the anchor chain, the following precautions shall be taken:

a) Use a wire rope preventer of adequate size to support the weight of the anchor to back up the housing stopper and prevent the accidental dropping of the anchor.

b) Use a chain stopper, if available; otherwise, insert a steel bar through a link across the upper lip of the chain pipe or lash with rope. Caution: Make sure the wildcat is engaged and the steel bar is removed before operating the wildcat.

c) Chain stoppers used for holding the anchor chain when used for towing shall be provided with a locking plate and cotter pin.

2.8.3 Emergency. Provisions such as large bolt cutters, cutting torch, release stopper, axe, or pelican hook shall be made for letting go of the towline in an emergency.

2.8.4 Turning. Build up turns slowly while getting under way. Never go from dead-in-the-water to a standard speed. Keep the tow abaft the beam.

2.8.5 Sinking. Disconnect the towline immediately if the tow is sinking. Failure to do so may result in damage or possible loss of the towing ship. Unless sinking of the tow appears imminent, do not abandon it. It is the towing ship's duty and responsibility to prevent the tow's loss.

2.8.6 Weather. Keep a lookout for small weather fronts; a sudden unexpected weather front can cause great damage. Never tow at short stay in rough weather. During favorable weather and seas, ensure that a sufficient catenary exists to absorb shock loading. Ensure that all rigging is adequate; if in question, always overrig.

2.8.7 Speed. Always be conscious of stresses in determining towing speed. The hull configuration, towing ship's horsepower, weather conditions, design of the rig, and other factors govern the towing speed. Towing shall be accomplished at the greatest speed possible without damage to the towing rig or tow.

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2.8.8 Monitoring the Tow. Monitor the tow during transit. Flooding alarms, draft marks, list, trim, and general seakeeping characteristics should be noted. Investigate any suspicious behavior or condition in the tow. On long transits, board an unmanned tow periodically for inspection.

2.8.9 Personnel Safety. Keep all unnecessary personnel from the vicinity of the towline. Personnel engaged in rigging or unrigging evolutions shall wear gloves, safety helmets, inherently buoyant life jackets, and safety shoes as required.

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Section 3: TOWED UNITS

3.1 Nonself-propelled Floating Craft. The service craft considered in this handbook may be towed in rivers, harbors, channels and the open sea. The types of nonself-propelled floating craft that have been the source of towing operations are listed in paras. 3.2 through 3.7.

3.2 Barges and Lighters. Barges and lighters designed for sea duty present no special towing problems. Towing of lighters not designed for sea duty should be limited to rivers and harbors. Arrangements can be made to transport them in floating drydocks as cargo for overseas assignments.

3.3 Cranes, Derricks, and Pile Drivers. These structures present towing problems peculiar to each type. It is frequently necessary to improvise protective devices, to provide internal structural reinforcement, and to add extra equipment to make them seaworthy. Consultation with NAVFACENGCOM, Code 04A3, is advisable for instructions and towing arrangement plans.

3.4 Dredges. When towing is contemplated, NAVFACENGCOM, Code 04A3, should be consulted concerning any special plans required for preparation of a tow. It is recommended that dredges of all types with attendant equipment be transported in floating drydocks.

3.5 Pontoons. Occasionally, circumstances require that Navy advanced base 5 ft x 7 ft x 5 ft (1.5 m x 2.1 m x 1.5 m) Navy Lighterage (NL) or International Organization of Standardization (ISO) container size pontoon gear be assembled into units and towed in the open sea. In some cases standard towing arrangement plans are available.

3.6 Graving Dock Caissons. Because of their box-like shapes, dock caissons, whether floating upright or flat, are unwieldy under tow, and are usually only towed once from the construction yard to the graving dock site. NAVFACENGCOM, Code 04A3, will assist contractor in preparation of special plans for any tow operation.

3.7 Offshore Structures. Elevated platforms or piers often are towed to the site and elevated into place. Special towing plans shall be prepared for each specific installation.

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Section 4: PREPARATION FOR TOWING

4.1 Preparation. The towed unit must be seaworthy and suitably outfitted prior to departure. OPNAVINST 4740.1C (3) specifies that the person who has cognizance of the towed units at the point of departure is responsible for the proper preparation of the towed unit for sea and the assembly of the required equipment. When preparing a crane, pile driver, dredge, dump scow, or other floating equipment designed for operation in sheltered waters, it may be necessary to remove high weights, secure booms, ladders, deck structures, ballast and trim, and perform other unique functions necessitated by the design of the hull, as illustrated in Figure 1.

4.2 Structural Design. Many of the units are structurally designed for towing in a seaway. Those that are not will require extensive bracing and local strengthening, or will require transport over the ocean as cargo in the well of a Dock Landing Ship (LSD) or other suitable vessel. Craft constructed of steel, timber and concrete, of various sizes and forms, are capable of safe towing over the sea with careful preparation and planning in regard to the involved risks of prolonged tows. Timber craft often require extensive bracing and local strengthening to resist the constant working action while being towed in a seaway (1).

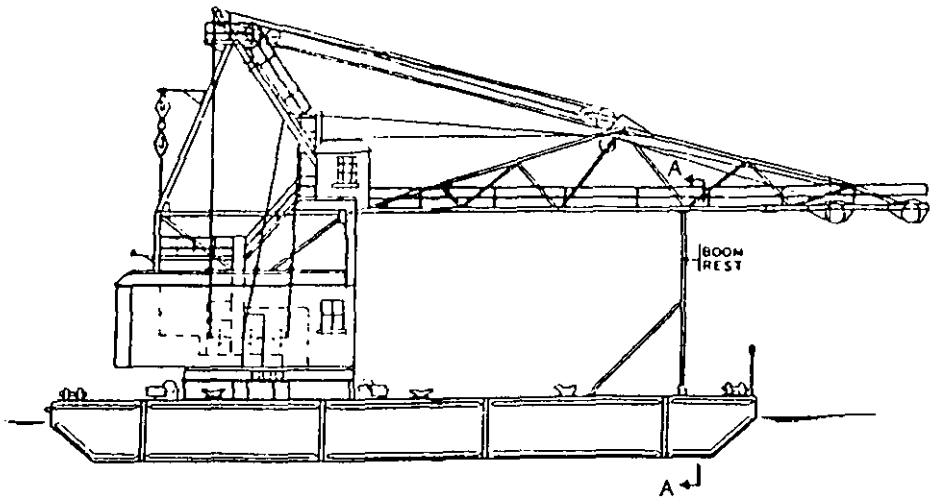
4.3 Seaworthiness. The craft to be towed shall be inspected to ascertain its material condition including potential weakness due to structural configurations and in accordance with the Material Inspection Survey Board requirements and procedures applicable to the craft under consideration. The types of floating craft within the scope of this handbook are not equipped for propulsion and have minimal or nonexistent utilities, habitability and navigational features, therefore, the inspection shall be primarily directed to the hull structure and watertightness areas for determination of satisfactory or unsatisfactory material conditions. All unsatisfactory items shall be repaired or corrected prior to departure to a state of readiness consistent with the characteristics of the ocean or coastwise tow to be accomplished.

4.4 Outfitting. An equipment inspection report form listing outfitting items ordinarily required is shown in Figure 2. Not all items listed are needed for every tow, and special equipment not listed may be desirable, therefore, some judgment is required in regard to equipment to be furnished for a particular tow. Towing gear arrangements and selection of ground tackle is discussed in Section 4. A decision regarding whether the tow will be manned or unmanned is a prerequisite to outfitting considerations.

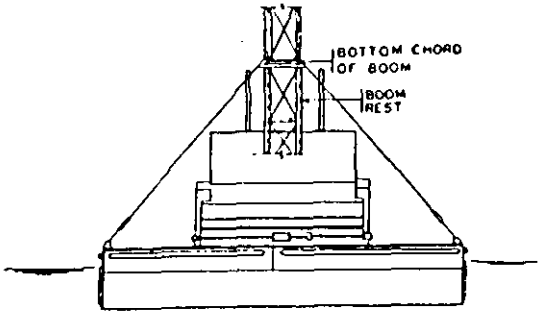
4.4.1 Manned Tows Outfitting. Shipboard systems are usually secured during an ocean tow. The equipment described in paras. 4.4.1.1 through 4.4.1.11 is normally provided:

4.4.1.1 Communications. Provide a short wave portable radio communications set with a working radius of about 50 mi. (80.4 km) for riding crew communications with towing vessel while at sea. Standard sets of code flags should be provided on the towed unit for visual communications.

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ELEVATION



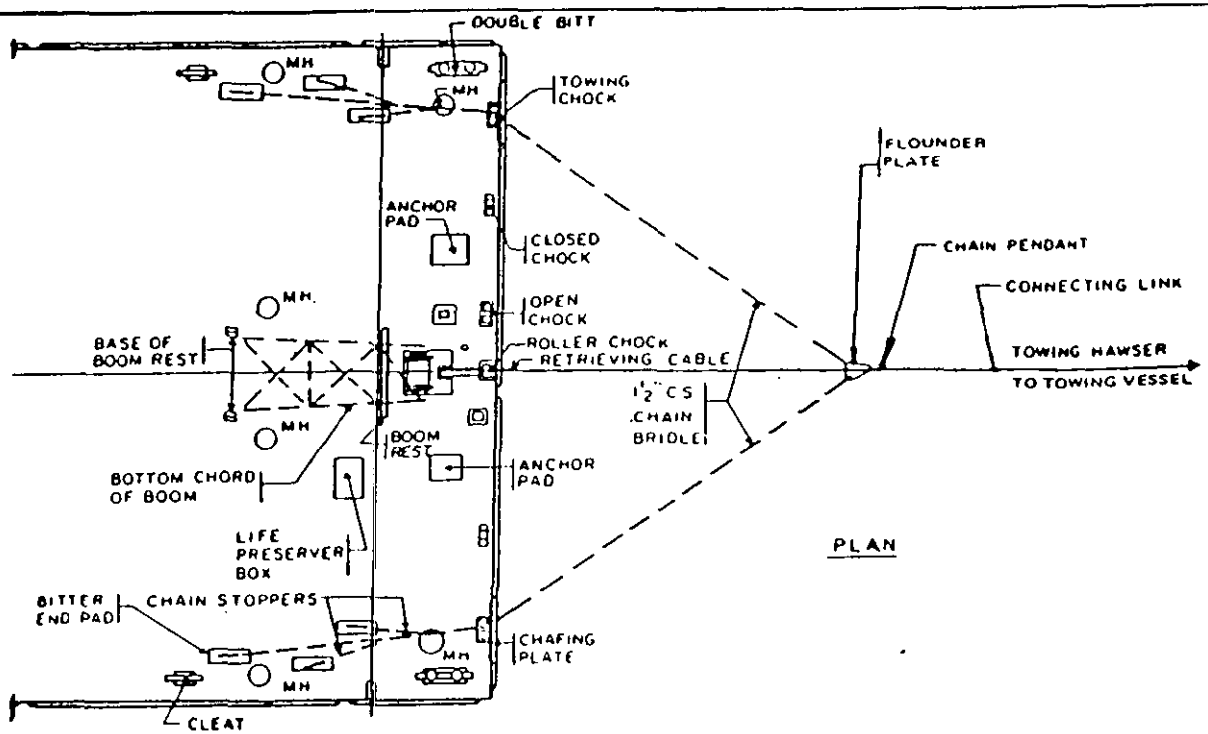
SECTION A-A

NOTES

1. Towing arrangement shown on these drawings is intended for use in inland or protected waters only.
2. Navigating and running lights to be provided.
3. If 1-1/2" chain bridle and fittings shown are not readily available, a wire rope bridle of equivalent strength may be used. All details and fittings to be revised accordingly.
4. On the above plan the bridle is shown at the bow end of the crane and the boom in a forward position. If the tow must pass through waters having strong following currents, the bridle can be installed at the stern (details same as shown on above plan) and the boom remain stowed in position shown.
5. The barge should be inspected thoroughly for water tightness. All vents to be sealed and necessary repairs made before departure. All equipment and fittings to be inspected and lashed or battened down securely where required.
6. The tow should be trimmed a total of one foot down by the trailing end. This trim to be provided by use of the existing water ballast tanks.

Figure 1
60-Ton Floating Crane - Towing Arrangement Plan & Details

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NOTES (con't.)

7. If it is known that the tow must pass through open and unprotected waters, where heavy rolling or pitching may occur, the crane must not be taken to sea unless the longitudinal and transverse stiffening trusses shown on Army Service Forces Dwg. No. 413B-1 are in place. If these trusses are found to be in place the above towing layout should be revised as follows for open sea towing:

   a. *Bridle as shown and noted is O.K. (Except wire rope should not be used in the bridle.)*

   b. Boom to be removed and stowed on main deck. It should be placed in wooden chocks, lashed down by wire rope or chain lashing with turnbuckles.

   c. The Machinery House should be made secure from rotation by using 8 1-1/2" tie rods and turnbuckles (2 at each corner) extending from machinery floor framing to barge deck. Locate tie rod padeyes on deck so as to provide 45° slopes on rods.

   d. All windows to be covered with 1" wood sheathing.

   e. Boom rest to be stowed on forward deck and lashed or clapped securely to deck.

   f. 1" chafing plate to be provided at open chocks as indicated on plan.

Figure 1 (Cont'd)
60-Ton Floating Crane - Towing Arrangement Plan & Details

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TOWING
INSPECTION REPORT
FOR
LIGHTERS, CRANES, DERRICKS, DREDGES, ETC.

NAME OR NUMBER _____ DATE _____

REPORTED BY _____ AT _____

TOWING GEAR

Pad eyes _____
Chocks _____
Rubbing strips _____
Bitts _____
Connecting links _____
Shackles _____
Chain _____
Wire _____
Chain stoppers _____
Spiders _____
Bridles _____
Pendant _____
Breaking lines _____
Reels _____
Retrieving gear _____
Windlasses _____
Capstans _____
Bow-wave bulwark _____
Stern-wave bulwark _____

ANCHOR AND GROUND TACKLE

Anchor and chain (fore) _____
Anchor and chain (aft) _____
Chain lockers _____
Shackles _____
Connecting links _____
Pad eyes _____
Chain stoppers _____
Breaking lines _____
Billboards _____
Securing gear _____
Engines _____
Chocks _____
Cleats _____
Rubbing strips _____
Bitts _____
Anchor davits _____

Figure 2
Sample Inspection Report Form (Obverse)

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HATCHES AND MANHOLES

Weather deck _____
Pontoon deck _____
Wing walls _____
Safety deck _____
Compartments _____

OTHER GEAR

Blocking _____
Cribbing _____
Bracing _____
Shoring _____
Lashings _____
Cargo _____

NAVIGATION

Lights _____
Instruments _____
Signaling _____
Distress _____
Charts _____
Leads _____
Logbooks _____

REMARKS

The above check-off list was completed by the undersigned representatives, who find that all work has been satisfactorily accomplished in accordance with the approved towing plans and confirm that the craft and towing equipment are considered seaworthy for the imminent towing operation.

_____
For Towing Officer

_____
For Towing Vessel

Figure 2 (Con't)
Sample Inspection Report Form (Reverse)

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4.4.1.2 Navigation lights. Towed units must be equipped with required navigational lights for vessels being towed in accordance with U. S. Coast Guard, Publication No. CG-169, Rules to Prevent Collisions of Vessels (6). In general, the following lights must be furnished as standard towing equipment: Side, Anchor, Towing, Not-Under-Command, and Man Overboard.

4.4.1.3 Lifeboats. Lifeboats or rafts and life vests are required equipment for manned tows.

4.4.1.4 Habitability. Riding crew accommodations would include quarters, galley and mess, fresh water, dry and refrigerated food, storage space, heating, sanitation, first aid, and recreation facilities. Experience and judgment will indicate the extent, location, and quantities of such logistics to be furnished on the towed unit utilizing existing spaces and installations.

4.4.1.5 Power. Electric power loads for a craft while under tow are much lower than the normal loads for which a shipboard power plant would be designed, therefore, it is generally more economical to install a portable diesel engine driven generator of the needed capacity. Electric loads could include power for the galley, refrigerator, heating, sanitation, lighting and pumps.

4.4.1.6 Pumps. To dewater bilge tanks or other compartments that may become flooded due to leakage or damage, provide pumping capability. It may be expedient to use portable gasoline power operated pumps or motor powered pumps if power is made available, or manually operated pumps for emergency use.

4.4.1.7 Fuels. Fuel oil storage in existing tanks, or properly stowed drums, will be required in quantities sufficient to supply the diesel-driven generator for the duration of the voyage. Storage for gasoline or similar fuel must be provided for gas-driven pumps, lifeboats or motor launches, or other uses as required, and stored in drums with provision for quick disposal in emergencies.

4.4.1.8 Fresh water. Potable water for cooking, bathing, and personnel use shall be stored aboard in existing or temporary storage tanks suitable for the purpose and adequately secured to the towed unit to preclude loss or damage during the voyage.

4.4.1.9 Fire protection. Provide dry chemical or foam fire extinguishers.

4.4.1.10 Crane storage. Cranes should be stowed in accordance with NAVFACENGCOM drawings. Secure and brace all parts to resist lateral and vertical movements.

4.4.1.11 Refrigeration. An emergency installation must be provided on the towing unit. Additional guidance can be found in (1).

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4.4.2 Unmanned Tows Outfitting. Outfitting of unmanned tows shall meet the following criteria:

a) Provide bridles at both ends of craft to facilitate pick-up.

b) Consider need for wave bulwarks of steel or timber to deflect seas coming over bow.

c) Provide navigation lights powered by batteries.

d) Make provision for boarding to examine and service craft.

e) Provide a flood alarm system. Refer to (3).

f) Consider provision of a collision bulkhead in bow section.

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Section 5: TOWING SYSTEM COMPONENTS

5.1 Arrangement Plan for Towing. An arrangement plan indicating towing components usually has been prepared for individual types of craft and should be obtained from the selected record plans aboard the craft or from NAVSEA. Such plans may be varied to suit local conditions if strengths or functions are not impaired in any way. Structures to be towed are, in general, outfitted to allow for single-line, bridle, abreast, or tandem towing. Any of these methods is satisfactory; the one used, however, shall be in general accordance with towing arrangement plans and as approved by the tug captain or towing officer.

5.2 Single-Line Towing. Accumulated towing data indicates this method, with a long towline, to be satisfactory for towing in the open sea. The towline usually consists of a plow-steel, wire-rope hawser connected to a length of chain called the pendant. The pendant is led through the bow chocks and secured to the bitter end pad on the towed unit. Chain stoppers are provided to give additional security and to allow for veering of the chain. Chafing of the pendant, reducing its strength, must be avoided as much as possible by utilizing chafing blocks or plates on the towed unit, occasionally veering the pendant, and greasing the wearing areas.

5.3 Bridle Towing. This method is the most satisfactory for towing large floating craft especially when close or congested waters such as rivers or harbors must be traversed. Figure 1 illustrates the arrangements for bridle towing of a floating crane. The towline consists of a plow-steel, wire-rope hawser shackled to a chain pendant. The chain pendant is shackled to a flounder plate at the forward end of the bridle. Each leg of the chain bridle is also shackled to the three-way link (flounder plate). The bridle legs pass through bow towing chocks and are shackled to the bitter end pads and chain stoppers. The spread of towing chocks determines the length of each bridle leg required to form an approximately equilateral triangle.

A bridle retrieving cable is connected to the three-way links. It leads through the center chock and is secured to the center towing pad. It is used to recover the bridle or veer the chains or pendants during long tows to minimize chafing action. The retrieving cable may be hauled in utilizing a recovery cable block and tackle arrangement and power by anchor windlass, capstan or power or hand winch. On small craft not equipped with power windlass or capstans, recovery may be by use of a block-and-tackle arrangement powered by a hand winch. The flounder plate is hauled in to repair a broken hawser or a broken bridle.

5.4 Abreast Towing. Because of increased hazards, towing abreast in a seaway is not recommended except for short distance tows during favorable weather.

5.5 Tandem Towing. Towing two or more craft in tandem is hazardous and, for important structures, should be used only in extreme emergencies. Special studies and analyses must be made for each tow. Multi-unit tows of any type should be avoided in restricted water areas or congested harbors or rivers.

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5.6 Ground Tackle. Ground tackle of the sizes and types suitable for normal mooring, or for use in an emergency, is essential for towing operations. The gear should be adequate for all conditions that may be encountered. If practicable, provide some means for retrieving the tackle.

5.7 Towing Vessel. Service craft considered in this handbook may be towed in rivers, harbors, channels, and the open sea. Only tugs especially built and fitted for open sea towing are adequate for moving units over long distances in a seaway. Many naval and cargo vessels are equipped with gear suitable for towing. The capability of a tug or towing vessel and the number of tugs for a tow is determined by the total resistance of the towed unit in relation to the towline pull (pounds per brake horsepower) of the tug. In general, a tug will be capable of producing 20 lb. pull/brake HP at slow speed and about 75 percent of that at a towing speed of about 8 knots. The towing vessel officers should be able to furnish the maximum towline pull versus speed for their vessel. Modern seagoing tugs are generally equipped with a special towing engine which stows the wire-rope towing hawser on drums and allows it to payout and haul-in automatically while maintaining a constant towing tension on the line, preventing propeller fouling, or overloading of the towing hawser.

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Section 6: TOWING SYSTEM LOADS

6.1 Towing Operation. Ground tackle of the size and type suitable and adequate for all conditions that may be encountered is essential for a towing operation. There are several resistance factors that tend to retard the forward motion of a towed unit and must be evaluated to determine the towline loads. These types of craft generally have blunt ends or low rake and are nearly box shaped. Data based on experiments on models and on actual conditions allow an estimate of resistance factor magnitudes for ground tackle selection as described in paras. 6.2 through 6.9.

6.2 Towing Resistance. Best estimates of towing resistance are developed from specific unit model test data for stillwater or seaway conditions, however, the following guidelines are presented to give the magnitude of these loads. The magnitude of frictional resistance caused by water friction acting on the moving unit's hull depends upon the unit's wetted surface area "S", the nature and condition of the hull surface, and the tow speed. The magnitude of the residual resistance caused by waves, eddies and wakes formed as the towed unit passes through the water depends greatly upon the unit's hull form.

Most of the towed units considered herein have blunt ends that reduce forward motion of a body passing through water by forming large waves, eddies and wakes. Eddies are masses of water in violent agitation, usually at the stern of a vessel; they have a holding action on forward movement. Resistance of this type mounts rapidly with speed. Wave-forming resistance is small at low speeds, however, with increase in speed, the water in front of the bow tends to roll up and forms waves more readily with blunt-end vessels than with vessels having ship-shaped bows. Resistance data for wave forming also depend on the area of the vessel below the water-line and its tow speed.

Total towing resistance is represented by the following formula:

EQUATION: $$R = (C_f + C_r) \times S \times (V_k \times 1.689)^2 \times K_1 \times K_2 \quad (1)$$

WHERE:

= Seawater Density (1.99 lb/sec^2/ft^4)

R = Towing Resistance (lb.)

V_k = Towing Speed (knots) 1 knot = 1.689 fps

S = Area of units wetted surface below the towing draft waterline (ft^2)

C_f = Friction resistance coefficient (dimensionless)
= 0.002 for clean hull surfaces
= 0.005 for hull surfaces heavily fouled by marine growth

C_r = Residual resistance coefficient (dimensionless)
= 0.003 for ship shaped hulls, V_k/L 0.3
= 0.015 for box shaped hulls, V_k/L 0.3
= 0.004 for ship shaped hulls, V_k/L 0.6
= 0.020 for box shaped hulls, V_k/L 0.6

L = Unit waterline length (ft)

K_1, K_2 = Resistance Factors (dimensionless)

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The resistance factors K_1 and K_2 are difficult to calculate. Guidance based on reasonable approximations, conservative judgment, and experience is given in the following sections. Additional resistance coefficient data is available in (1), (3), Principles of Naval Architecture (4), and Recent Developments in Barge Design (5).

6.3 Towing Speed and Seaway Augment to Towing Resistance. Waves and resulting vessel motions such as heave, surge, sway, pitch, roll and yaw, can cause towing resistance to increase as much as 40 percent in a moderate to rough seaway if towing speed is sustained. In calm seas, tow speed normally should not exceed 8 knots. While in a moderate seaway, tow speed generally should be reduced to at least 25 percent to avoid increased loads and damage to the towed unit and towing system. A load factor (K_1 = 1.4) should be applied to towing resistance, Equation (1), to provide a seaway load margin when operating in locations where seaways can develop to 8 ft wave heights. Refer to Table 1, and (1), Table 3 and 4. Seaway effects are described in further detail in (3).

6.4 Shallow Water Augment to Towing Resistance. Towing resistance can increase by more than 20 percent at normal tow speed in shallow, restricted waters where the unit's draft is in excess of 70 percent of water depth. The unit's tow speed normally should be reduced at least 25 percent when operating in restricted waters, however, a load factor (K_1 = 1.2) should be applied to tow system design as a shallow water towing resistance margin. Either the seaway augment or the shallow water augment load factor is applied in Equation (1), but not both since a unit is not likely to be towed where both conditions exist. Shallow water effects are further described in (3) and (4).

6.5 Towline Springing and Drag Loads. Under certain sea conditions the towline may spring clear of the water surface causing shock and vibration affecting the load in the towline.

The towline dragging through the water and in some cases sliding on the bottom occurs most frequently when towing in rivers, channels, and harbors. Towline drag increases appreciably when towed units change course or drift from a following course which causes the line to form a large loop that passes laterally through the water. The load in the towline can increase by as much as 50 percent under the combined effect of towline springing or drag so that a load factor (K_2 = 1.5) should be applied to Equation (1) as a load margin.

6.6 Vessel List and Trim. When properly prepared for tow, floating structures will have no list. A 3- to 5 ft (0.9 m to 1.5 m) trim-down by the stern is desirable for large floating craft under tow. Flooding the aft ballast tanks is the preferred method for trimming-down a towed vessel by the stern. Heavy equipment or cargo positioned aft of midship is another satisfactory method for producing trim-down by the stern.

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Table 1
Nomenclature for Sea Conditions,
Swell Conditions, and Wind Force Conditions

Code Figure	Approx height of Sea, feet	SEA CONDITIONS *Seaman's description*
0	0	CALM, Sea like mirror.
1	less than 1	SMOOTH. Small wavelets or ripples with the appearance of scales but without crests.
2	1 to 3	SLIGHT. The waves or small rollers are short and more pronounced; when capping, the foam is white but has more of a glassy appearance.
3	3 to 5	MODERATE. The waves or large rollers become longer and begin to show whitecaps occasionally. The sea produces short rustling sounds.
4	5 to 8	ROUGH. Medium waves that take a more pronounced long form with extensive witecapping and white foam crests. The noise of the sea is like a dull murmur.
5	8 to 12	VERY ROUGH. The medium waves become larger and begin to heap up, the whitecapping is continuous, and the seas break occasionally. The foam from the capping and breaking waves begins to be blown along in the direction of the wind. The breaking and capping seas produce a perpetual murmur.
6	12 to 20	HIGH. Heavy, whitecapped waves that show a visible increase in height and are breaking extensively. The foam is blown in dense streaks along in the direction of the wind. The sea begins to roll and the noise of the breaking seas is like a dull roar, audible at greater distance.
7	20 to 40	VERY HIGH. High heavy waves developed with long overhanging crests that are breaking continuously, with a perpetual roaring noise. The whole surface of the sea takes on a white appearance from the great amount of foam being blown along with the wind. The rolling of the seas becomes heavy and shock-like.
8	40 and over	MOUNTAINOUS. The heavy waves becomes so high that ships within close distances drop so low in the waves troughs that for a time they are lost from view. The rolling of the sea become tumultuous. The wind beats the breaking edge of the seas into a froth, and the whole sea is covered with dense streaks of foam being carried along with the wind. Owing to the violence of the wind, the air is so filled with foam and spray that relatively close objects are no longer visible.

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Table 1 (Cont'd)
Nomenclature for Sea Conditions,
Swell Conditions, and Wind Force Conditions

Force of Wind	WIND FORCE CONDITIONS	Velocity	
Code Figure	Beaufort's Scale	Velocity, Knots	Weather Bureau Forecasts
0	CALM, Sea like a mirror.	less than 1	Calm
1	LIGHT AIR. Ripples, no foam crests.	1 to 3	Light
2	LIGHT BREEZE. Small wavelets, crests have a glassy appearance and do not break.	4 to 6	Light
3	GENTLE BREEZE. Large wavelets, crests begin to break; scattered whitecaps.	7 to 10	Gentle
4	MODERATE BREEZE. Small waves becoming longer; frequent white caps.	11 to 16	Moderate
5	FRESH BREEZE. Moderate waves taking a more pronounced long form; many whitecaps, some spray.	17 to 21	Fresh
6	STRONG BREEZE. Large waves begin to form; extensive whitecaps everywhere, some spray.	22 to 27	Strong
7	MODERATE GALE. Sea heaps up and white foam from breaing waves begins to be blown in streaks along the direction of the wind.	28 to 33	Strong
8	FRESH GALE. Moderately high waves of greater length; edges of crests break spindrift. The foam is blown in well-marked streaks along the direction of the wind.	34 to 40	Strong
9	STRONG GALE. High waves; dense streaks of foam along the direction of the wind. Spray may affect visibility. Sea begins to roll.	41 to 47	Gale
10	WHOLE GALE. Very high waves; the surface of the sea take on a white appearance; the rolling of the sea becomes heavy and shocklike. Visibility is affected.	48 to 55	Gale
1	STORM. Exceptionally high waves. Small and medium sized ships are lost to view for long periods.	56 to 63	Whole Gale
2	HURRICANE. The air is filled with foam and spray; sea completely white with driving spray; visibility very seriously affected.	64 to 71	Hurricane
3	HURRICANE.	72 to 81	Hurricane
4	HURRICANE.	81 to 89	Hurricane
5	HURRICANE.	90 to 99	Hurricane
6	HURRICANE.	100 to 109	Hurricane
7	HURRICANE.	110 to 118	Hurricane

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Table 1 (Cont'd)
Nomenclature for Sea Conditions,
Swell Conditions, and Wind Force Conditions

Code figure	Approx. height feet	SWELL CONDITIONS Description		Approx. length, feet
0	0	No swell		0
1	1 to 6	Low swell	Short or Average	0 to 600
2			Long	Above 600
3	6 to 12	Moderate	Short	0 to 300
4			Average	300 to 600
5			Long	Above 600
6	Greater than 12	High	Short	0 to 300
7			Average	300 to 600
8			Long	Above 600
9	.....	Confused		

The direction from which the swell moves to be recorded in numerical points from 1, North by East, to 32 degrees North.

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6.7 Wind Resistance. Wind force is particularly great on floating structures having a large sail area, especially during a violent storm. Wind resistance may be so great that the towed unit cannot be moved forward by the tug, and may possibly become extremely difficult to control at all. Resistance data for wind resistance depend on the area of vessel above the waterline subjected to wind, wind velocity, tow speed, and form of vessel superstructure as represented by the equation:

EQUATION: $W = 0.004 (V_w + V_k)^2 \times C \times f_3$ (2)

WHERE:
- W = Resistance (lb.).
- C = Cross-sectional area vessel above the water line (ft^2)
- V_w = Wind velocity (knots).
- V_k = Tow speed relative to still water (knots).
- f_3 = Coefficient depending on the form of vessel subjected to wind. Use 0.6 for faired forms to 0.7 flat forms exposed to the wind.

6.8 Factor of Safety and Total Design Load. The sum of Equation (1) and Equation (2) should be multiplied by a safety factor to determine total towline design working load as follows:

TOWING MODE	MINIMUM SAFETY FACTOR
Towing engine and chain pendant	1.5
Chain pendant without towing engine	2.0
On Bitts, Padeyes, etc. (without pendant or towing engine)	3.0

6.9 Analytical Solutions for Towline Tensions. Analytical solutions can be found for predicting towline tensions in the coupled dynamic system consisting of towing vessel, towline and towed vessel operating in a seaway. The Naval Systems Engineering Command (NAVSEC) Code 56W-23 or commercial naval architecture firms have computer codes which can be used to solve for loads in parts of the system, and with additional development, loads in the entire system.

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Section 7: TOWING SYSTEM COMPONENT SELECTION

7.1 Systems Components. Towing system components are selected to withstand the loads previously described with a reasonable margin of safety as well as to provide an effective system which is easily deployed, operated and maintained. Detailed guidance and design data is available in (1) and Modern Seamanship (7), and in paras. 7.2 and 7.3.

7.2 Towlines. The three kinds of towlines used extensively are:

a) Manila rope - manila rope is satisfactory for towing small craft. When of sufficient length, this rope is heavy enough to form a satisfactory towing catenary, yet not too heavy for easy handling or stowing.

b) Synthetic Fiber Rope - synthetic fiber rope such as nylon and polyester in short pendant or longer hawser sections provides energy absorption to the towing system without increasing its catenary and is light and easy to handle. More detailed guidance in its use is available in (1) and NAVFAC P-401, Pontoon System Manual (8).

c) Wire Rope - Wire rope is the most satisfactory towing hawser for the heavier vessels and is especially well suited for towing over long distances. The advantages of wire rope are its great strength, its flexibility, and relatively long life.

7.2.1 Towing Hawsers. Towing hawsers are tug furnished equipment and selection is made by the towing officer in relation to the tugs to be used. For general guidance, the working stress of a towing hawser must not be greater than 60 percent of the breaking stress when applied to the total resistance. Type E, 6 x 37 high grade plow steel, galvanized, fiber core, wire rope is a suitable type of hawser material. Characteristics of natural fiber, synthetic fiber, and wire ropes are presented in NAVFAC DM-26.6, Mooring Design Physical and Empirical Data.

7.2.2 Towline Lengths. The towline lengths between tugs and towed units are varied to best suit the sea and wind conditions being encountered. The towline lengths shall be reduced in shallow waters to prevent dragging on the bottom. A 300-fathom towline is considered adequate for most sea operations. Normally, approximately 200 fathoms of line are payed out for constant use, and 100 fathoms held in reserve on the towing engine drum for adjustment when needed. If possible, vary the length of towline to enable a tug to ride the crests and troughs of waves simultaneously with the towed unit. When properly adjusted, the control of towing operations should be comparatively easy and steady.

The uniformity of wave lengths in a seaway at a given time helps in controlling a tow for that particular period. Constant control during a voyage is not advisable because of the difficulty of changing towline lengths to conform to periodic changes in wave length. Control should be maintained during rough weather when circumstances allow. If a length of line is such that the towed vessel is in a wave while the tug is riding a crest, the line

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will slacken and then tighten with sudden jerks. In heavy weather, a jerking line is more likely to fail. Each line should be fitted with a standard steel thimble or socket on the end shackled to the pendant. The other end should be plain and attached to the bitter end fitting on a towing engine drum.

7.3 Towing Pendants. When towing with a bridle, the pendant is that portion of the towline shackled between the flounder plate and wire-rope hawser. A pendant usually consists of 15 to 30 fathoms of chain. It adds weight to a line causing a towing catenary of reasonable depth to be formed. For single-line towing, a longer chain pendant is used instead of a bridle. The working stress of the pendant and bridle chain must not be greater than 60 percent of the breaking stress, but the stress used after increasing the resistance by the factor of safety above must be less than the proof stress, but in no case be more than 70 percent of the breaking stress.

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Section 8: TOWING RECORDS

8.1 Recordkeeping. Towed units receive severe service under tow and their performance must be recorded on special reports provided for this purpose. These reports are to be completed before, during, and after each tow with a copy of each retained in the service craft history file.

8.2 Inspection Report. Figure 2 has been prepared for recording the towing gear and ground tackle installed for the tow, the security of hatches and manholes, descriptions of cargo lashings and blocking, and installations of navigational equipment. The form is a general check list for the items required for towing and only the items applicable to the tow at hand need be listed. It is important that gear and structure be inspected and a tow should not depart until it has been judged seaworthy and the towing inspection report has been signed by the officials designated thereon.

8.3 Description Report. Figure 3 has been prepared to record data about the towed unit and towing vessel. The description of specific characteristics will assist analysis of towing failures, and provide experience gained from each tow for future designs and preparation of towing plans.

8.4 Behavior Report. Figure 4 has been prepared to record the dynamic rolling experienced by the towed unit throughout the tow period. Designers desire this information for designing foundations, lashings for cargoes, stability analyses, and other items for securing equipment to the towed unit. Use Standard Defense Mapping Agency, Hydrographic/Topographic Center (DMAHTC) nomenclature in reporting sea and swell conditions, and wind force, as described in Table 1.

8.5 Performance Report. Figure 5 has been prepared as a guide to record data about towed unit and environmental conditions as well as voyage distance and speed. This data will aid in planning future voyage fuel and power requirements as well as provide valuable information for future designs of tugs and towing systems.

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TOWING
DESCRIPTION REPORT
FOR
LIGHTERS, CRANES, DERRICKS, DREDGES, ETC.

NAME OR NUMBER _____ DATE _____

REPORTED BY _____ AT _____

DESCRIPTION OF TOWED UNIT

Draft at departure _____ Fore _____ Aft

Draft on arrival _____ Fore _____ Aft

Cargo aboard at departure _____

Cargo aboard on arrival _____

Depth of water in each ballast tank at departure

Tank No.	Water Depth	Tank No.	Water Depth	Tank No.	Water Depth	Tank No.	Water Depth
1	_____	4	_____	7	_____	10	_____
2	_____	5	_____	8	_____	11	_____
3	_____	6	_____	9	_____	12	_____

Depth of water in each ballast tank on arrival

Tank No.	Water Depth	Tank No.	Water Depth	Tank No.	Water Depth	Tank No.	Water Depth
1	_____	4	_____	7	_____	10	_____
2	_____	5	_____	8	_____	11	_____
3	_____	6	_____	9	_____	12	_____

Crew complement _____ Officers _____ Men _____

SAMPLE

Figure 3
Sample Desciption Report Form (Obverse)

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DESCRIPTION OF TOWING UNITS

MAIN VESSEL

Name _____
Type _____
Shaft hp _____
Fuel consumed _____
Towing engine _____
_____
Hawser, size _____
length _____
Pendant, size _____
length _____
Bridle _____

ASSISTING VESSEL

Name _____
Type _____
Shaft hp _____
Fuel consumed _____
Towing engine _____
_____
Hawser, size _____
length _____
Pendant, size _____
length _____
Bridle _____

AUXILIARY VESSEL

Name _____
Type _____
Shaft hp _____
Fuel consumed _____
Towing engine _____
_____
Hawser, size _____
length _____
Pendant, size _____
length _____
Bridle _____

STAND-BY VESSEL

Name _____
Type _____
Shaft hp _____
Fuel consumed _____
Towing engine _____
_____
Hawser, size _____
length _____
Pendant, size _____
length _____
Bridle _____

DESCRIPTION OF VOYAGE

Departure from _____ Date _____ Time _____

Destination _____ Date _____ Time _____

Distance _____ Average Speed _____ Maximum speed _____

Voyage information not covered above:

SAMPLE

Figure 3 (Cont'd)
Sample Description Report Form (Reverse)

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TOWING
BEHAVIOR REPORT
FOR
LIGHTERS, CRANES, DERRICKS, DREDGES, ETC.

ON TOW FROM _____ TO _____

NAME OR NUMBER _____ DATE _____

REPORTED BY _____ AT _____

PERIOD AND ANGLE OF ROLL

(Obtain data by most practical and accurate means available, using stop watches, gyro list indicators, level indicators, etc.)

Time of Reading	Sea Conditions	Force of Wind	Swell Conditions	Roll to Starboard	Total Roll	Time of Roll

SAMPLE

(Use other side if necessary)

Figure 4
Sample Behavior Report Form

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PERFORMANCE REPORT FORM														
Month	Towed unit				Place of Departure	Destination	Distance nautical miles	Condition		Speed, knots	Towing vessel		Type of tow	Remarks
	Craft number	Mean draft	Trim, ft.	Crew				Sea	Wind		Number	Fuel per nautical miles, gal		

Figure 5
Performance Report Form

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9. Navy Safety Precautions for Forces Afloat, OPNAVINST 5100.19A, Articles 0212 and 0281, Department of the Navy, Office of Naval Operations.

10. NAVFACENGCOM Design Manuals
NAVFAC DM-26.6 Mooring Design Physical and Empirical Data

Government agencies may obtain Design Manuals from the U.S. Naval Publications and Forms Center, 5801 Tabor Ave., Philadelphia, PA 19120. TWX: 710-670-1685, AUTOVON: 442-3321. The stock number is necessary for ordering these documents and should be requested from the NAVFACENGCOM Division in your area.

Non-Government organizations may obtain Design Manuals from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

CUSTODIAN
NAVY - YD

PREPARING ACTIVITY
NAVY - YD

PROJECT NO.
FACR-0209

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-HDBK-1038/5	Towing Nonself-Propelled Floating Structures

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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